

LPG, fuel cells, ethanol, bio-diesel, etc? The reason is that combustion engines are highly inefficient as they run at only 20-25% efficiency in best-case scenario; electric drives operate at over 90% peak efficiency. Also, if the electricity is generated from renewable energy sources such as solar, wind, or hydro, there is zero emission.

Besides, the automotive industry is moving away from one-size-fits-all to more nimble, need-based and customisable product offerings. Different people have different requirements, and all want efficient solutions. Battery prices are coming down as well, making the financial equations tilt even further in favour of EVs.

One of the most important aspects that is fueling the growth of EVs is their simplicity. A typical combustion engine based car has about 1,700 moving parts. On the other hand, EVs have only 20 or less moving parts, which make them simple. We should focus on simplicity as long as it does not compromise performance or any usage scenarios.

Global context

The move towards electric mobility has already started globally. The sales of EVs have been accelerating at more than 40% year-to-year this decade, and EVs accounted for 2.6% of global vehicle sales in 2019. The number of public charging locations have doubled within two years to facilitate the increasing demand for EV charging infrastructure. Moreover, leading OEMs have announced EV portfolios up to 25% in their sales mix for the next decade. Volkswagen, BMW, and Daimler are prominent among them, indicating that the EV market is picking up.

Fueling this growth is the rapid decline in battery prices. High upfront price of batteries has been one of the major causes of high vehicle prices, which led to low penetration. Batteries are about 35-40% the cost of an EV, and the point when it becomes convenient and relevant for industries and consumers is when the prices are close to or less than \$100/kWh.

As a result of this growth, all OEMs are doing some R&D activities related to electric mobility, as it is the only

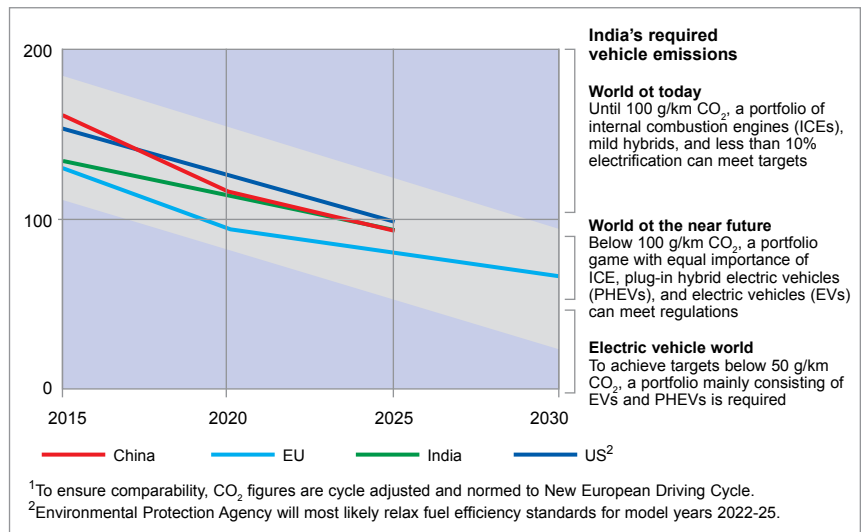


Fig. 1: Change in regulation of grams per kilometer (g/km) of CO₂¹ (Index 2004=100) (Source: International Council on Clean Transportation; McKinsey analysis)

way to reduce emissions. The norms will also hit India and ensure that all automotive brands have some xEVs (hybrid or full electric vehicles) as a part of their portfolio.

But what does it take for disruption?

We are seeing early signs of adoption and a steady growth. But disruption is different. It refers to major changes occurring very rapidly. One example of disruption is that which has been brought about by mobile phones, making the PCOs redundant.

Disruptions happen often. We have seen the drastic change in mode of transportation brought about by Uber in just one decade, especially in major cities like New York. But what would it take to disrupt the market and accelerate the adoption of EVs in India?

Vehicle + Commuting for EV adoption

India has had EVs in the automotive industry for a very long time. The first EV hit the market back in 2001. But the adoption is still slow. This is because when someone buys a vehicle, it is not just the vehicle consumers look at for overall suitability. Parameters like predictability of the route, utilisation of the vehicle, price difference between a combustion engine vehicle and electric vehicle, emerging technologies being adopted by the OEMs, government policies, etc, all play an important role.

When we classify different kinds of vehicles according to above parameters, we observe that the number one market that we are ready to serve is going to be of commercial vehicles, largely 3-wheelers and 2-wheelers. That is why last mile mobility is getting the kind of traction it is getting.

When we analyse from an energy parity perspective, regardless of electric or field based energy, we start seeing parity in battery prices around 200-250 dollars. EVs will find a lot of suitability if they have the right kind of life cycle. The moment that happens, ownership cost will reduce, ownership will become easy, and hence EVs will start getting adopted.

Trends suggest (see Fig. 1) that an explosion is going to happen in terms of 2- and 3-wheeler EVs in Indian markets by 2025. The push is in the right direction because the areas and the aspects which will help in EV adoption are the ones that we see all the time. Hence the disruption!

Emerging markets: A different reality

India and other emerging markets have a very different reality. When we design EVs for India, we have to keep the Indian realities in mind. We have densely populated roads with heterogeneous traffic and no uniformity in speed. Potholes and waterlogged roads are common in monsoon. Also, overloading is a common practice in commercial vehicles.